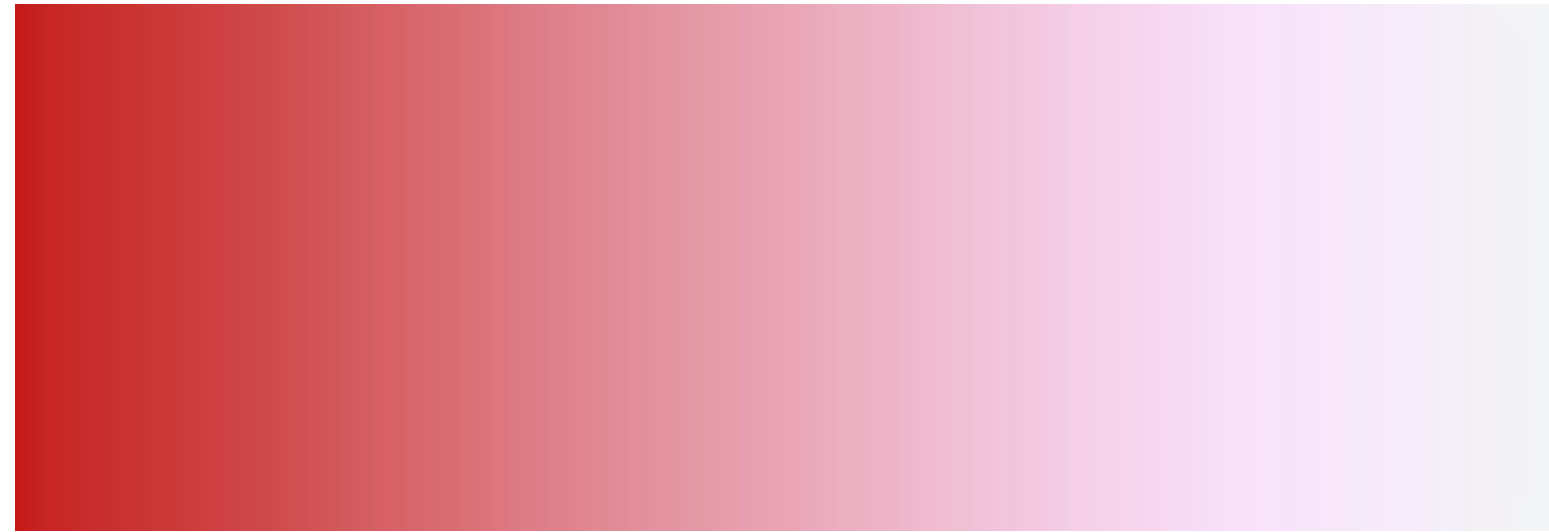


PBS Evaluation

Clinical Principle & Diagnostic Significance



**Low Power
(100x Total)**

Medium: Air



**High Dry
(400x Total)**

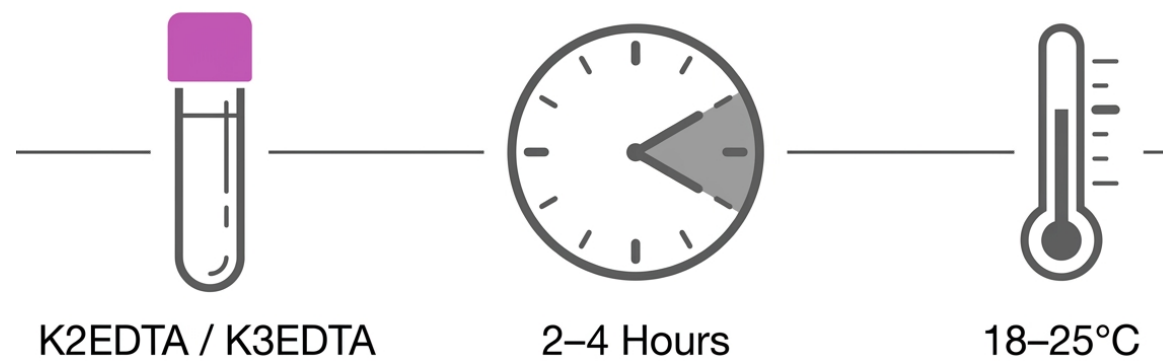
Medium: Air



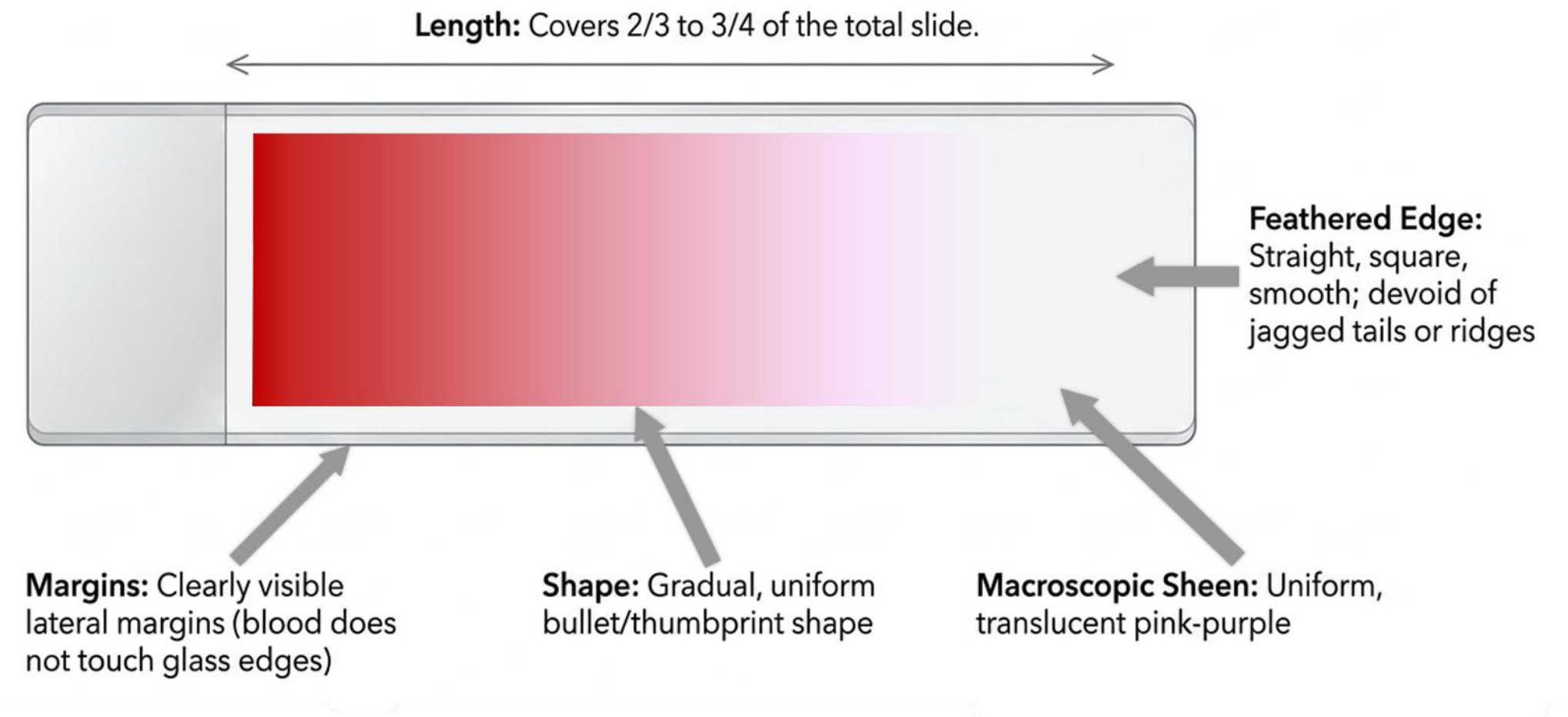
**Oil Immersion
(1000x Total)**

Medium: Type A/B Oil

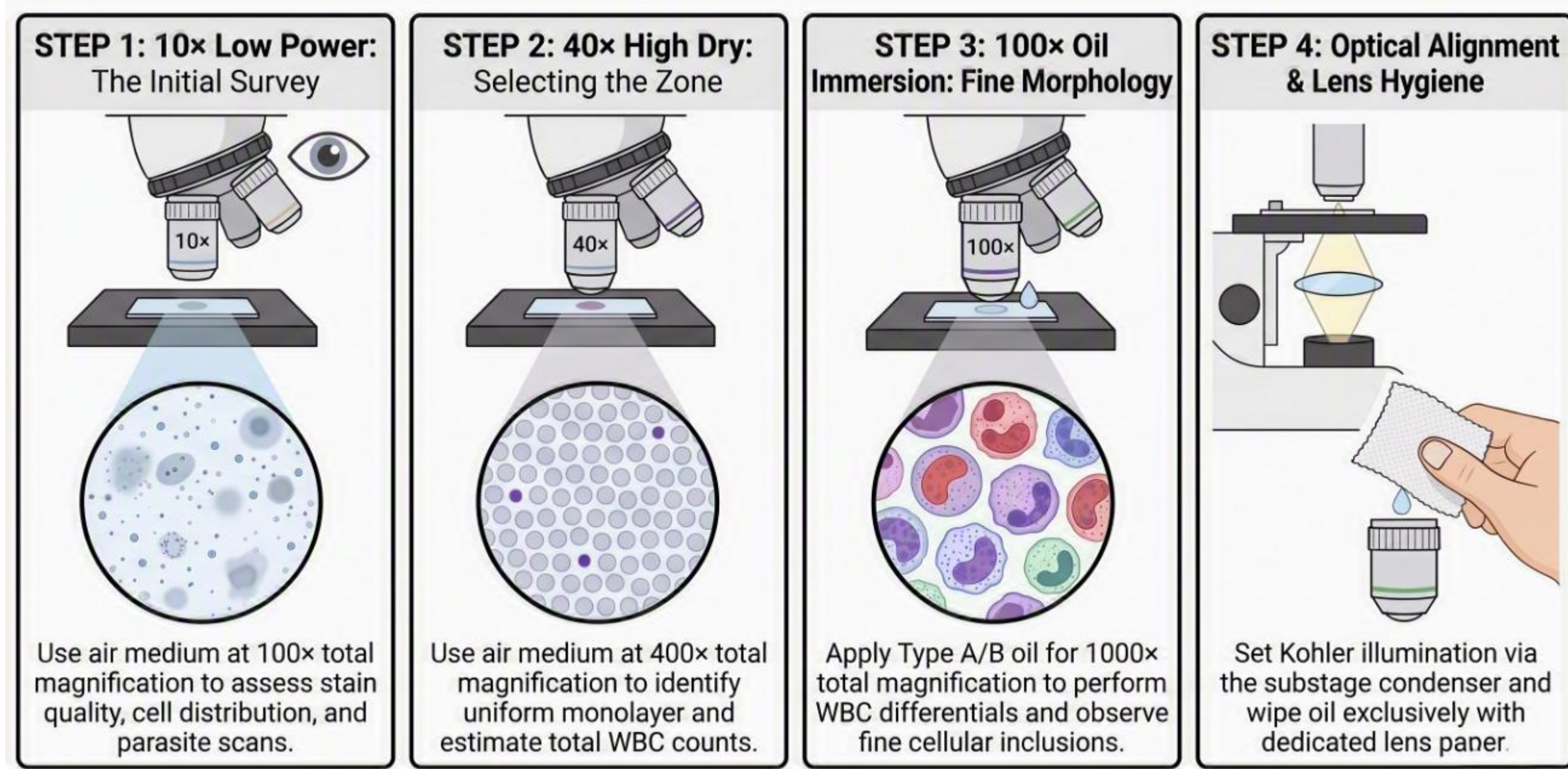
Specimen Requirements & Smear Integrity



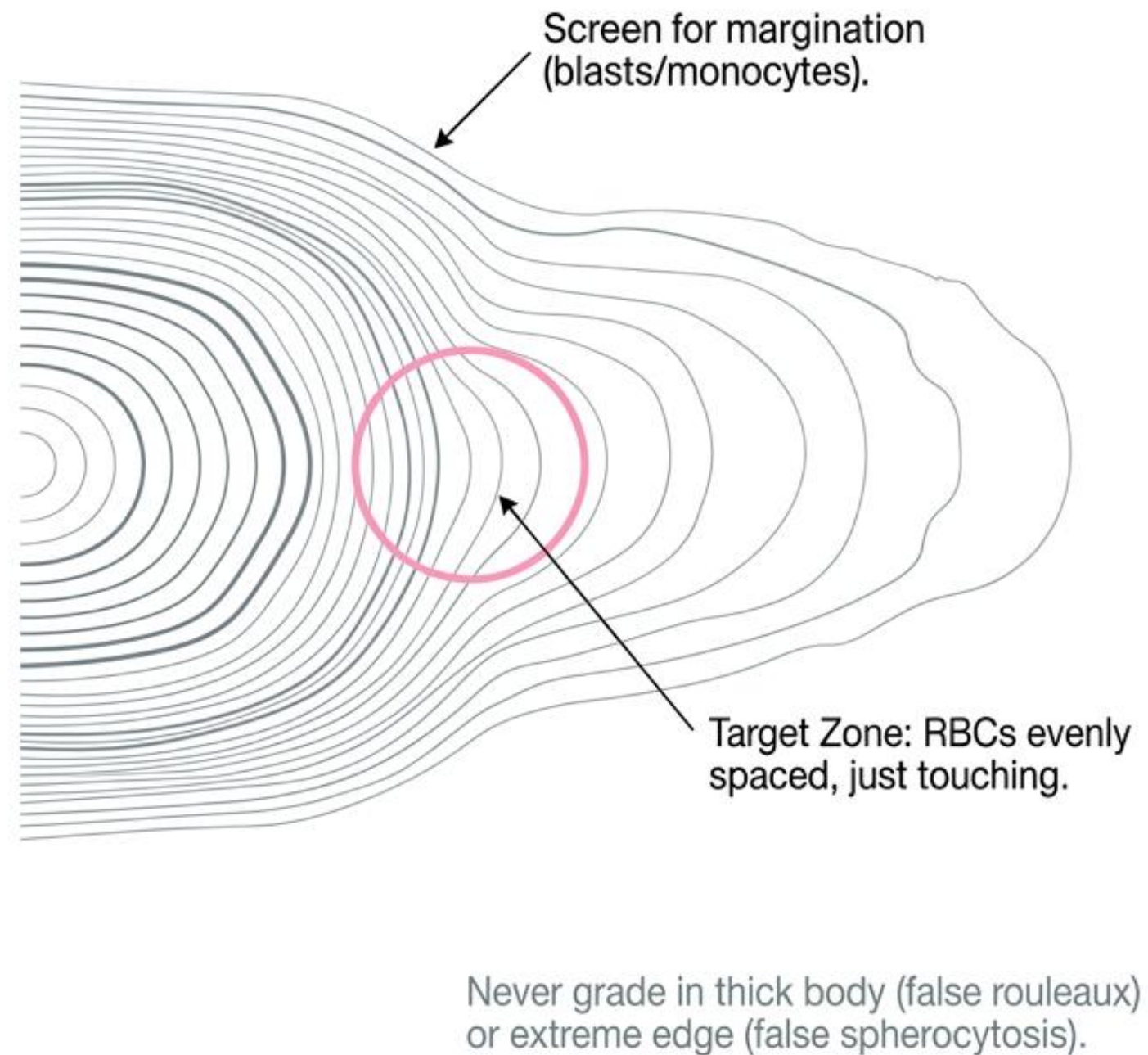
⚠ >4h induces necrobiosis, vacuolization, and platelet swelling.



Microscopic Equipment & Optical Protocol



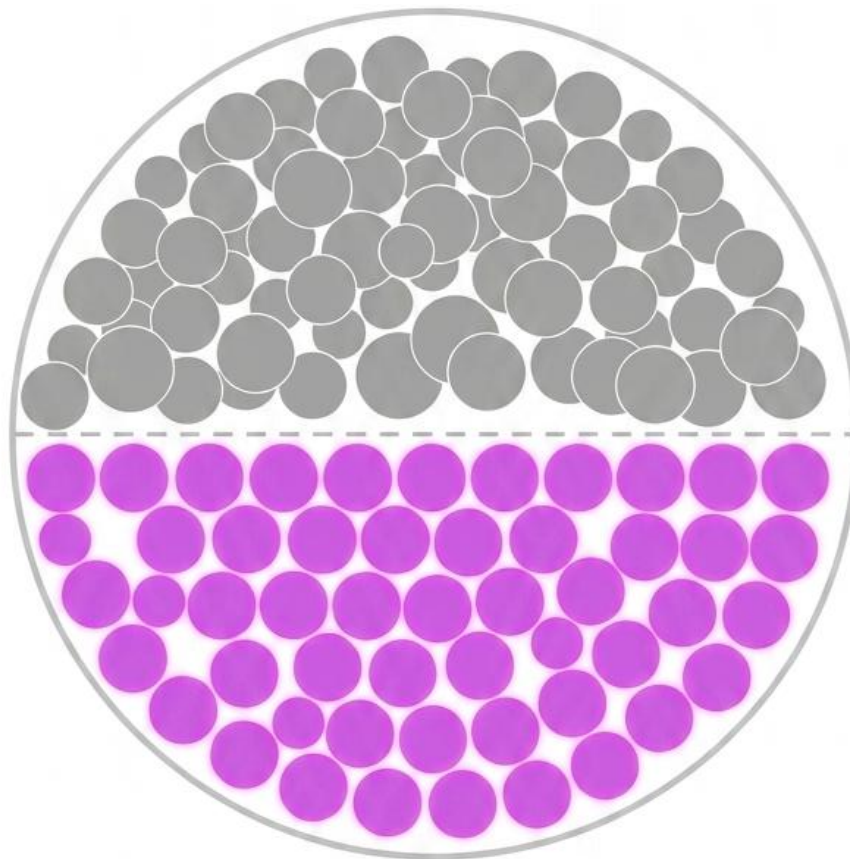
Low-Power (10×)



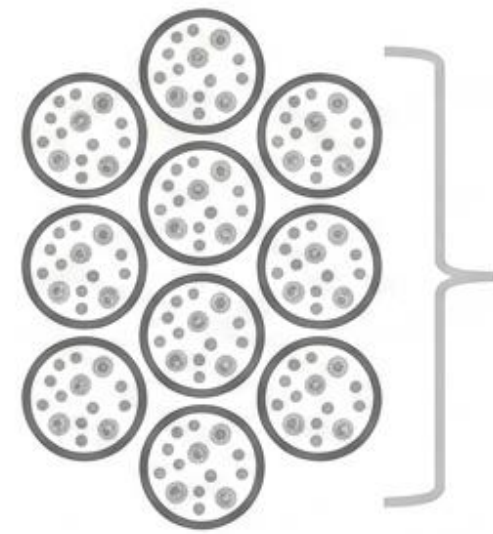
-  Leukocyte margination (Rogue blasts)
-  Fibrin microclots
-  Platelet clumps (Pseudo-thrombocytopenia)

High-Dry (40×)

Thick Body
(Artifactual
rouleaux /
microcytosis)



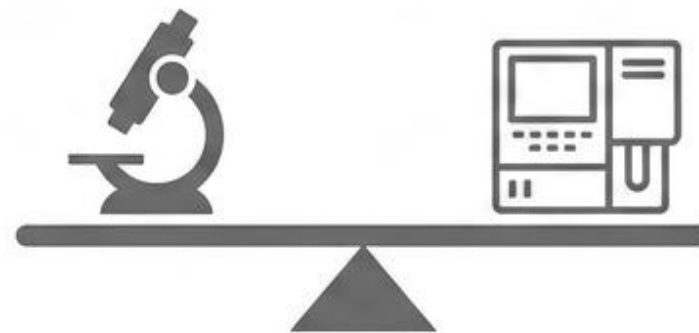
Monolayer
(Erythrocytes
just touching)



$$\div 10 \times 2,000 =$$

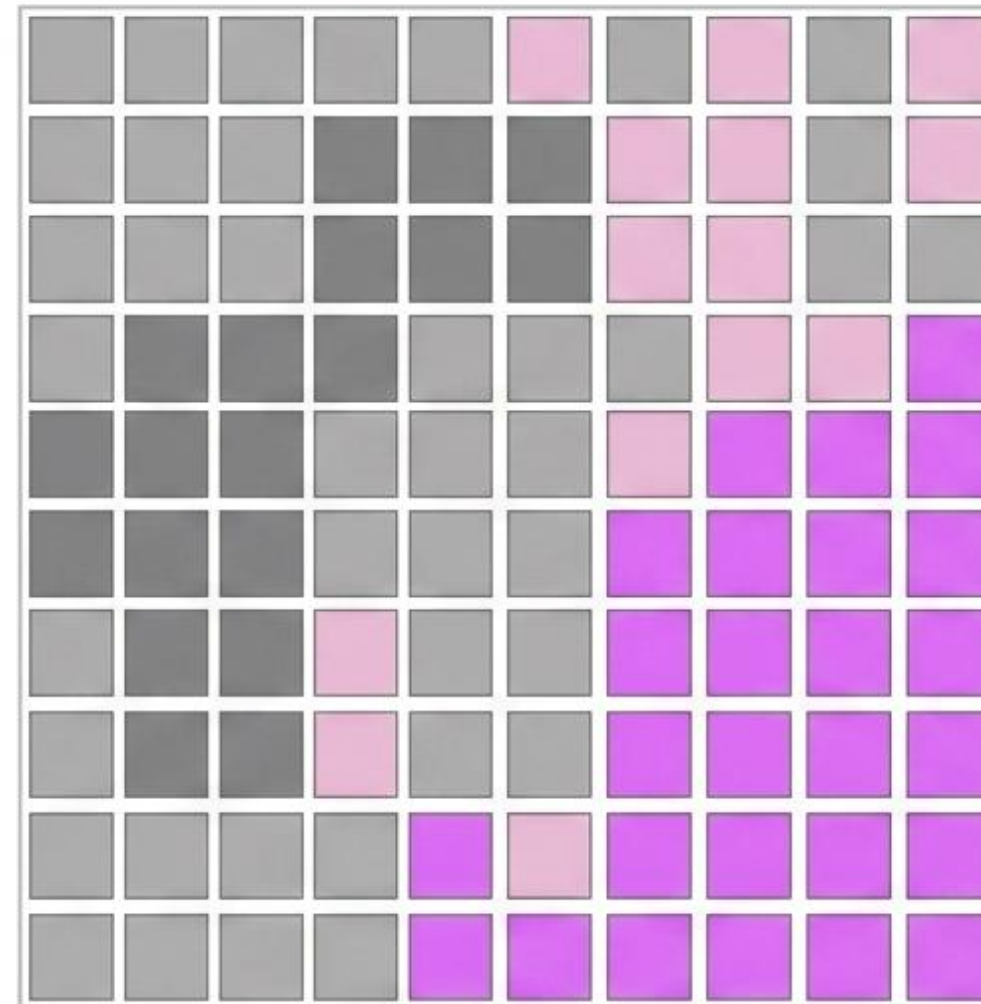
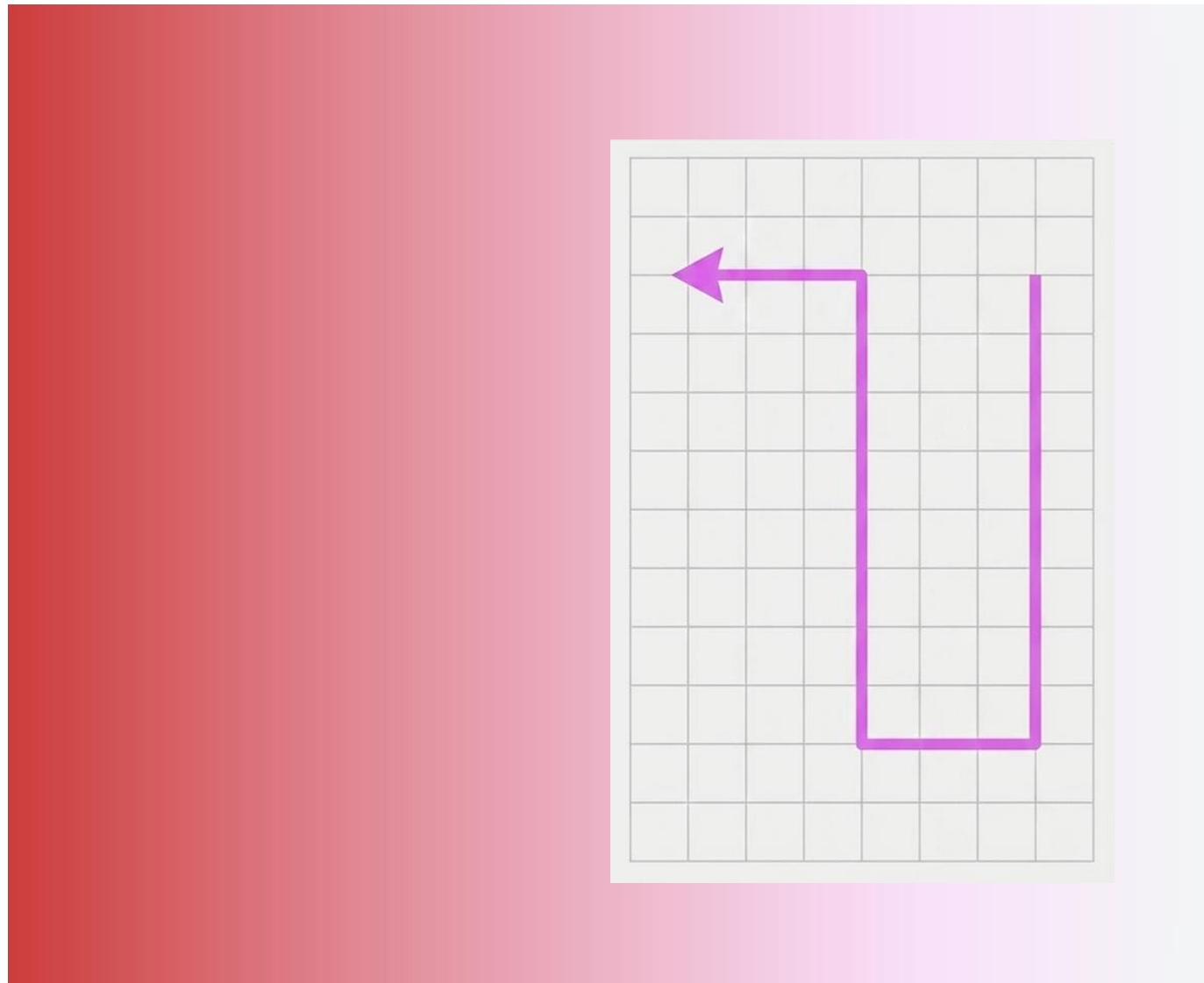


$$\text{WBC}(10 \text{ HPF}) \times 2,000 = \text{Estimated cells}/\mu\text{L}$$



Validate: Must agree with
automated analyzer within $\pm 20\%$

Oil Immersion (100×)



Classify 100 consecutive nucleated leukocytes.



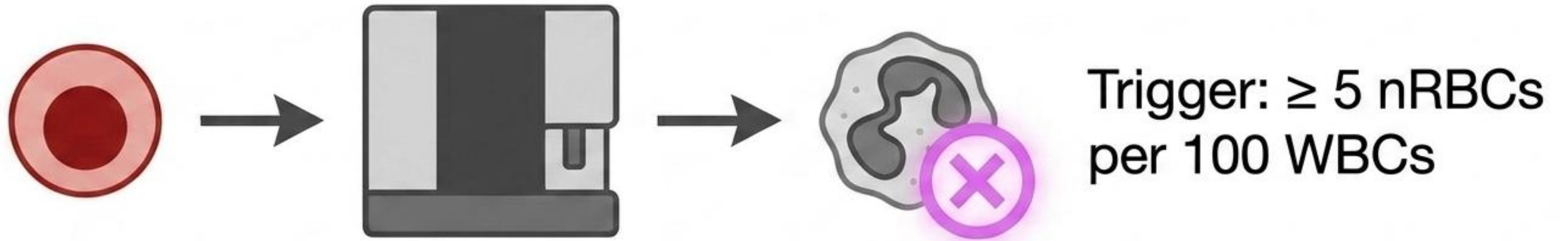
Exclude:
nRBCs and
Smudge cells
(tabulate
separately).

Platelet Estimation


$$\left[\begin{array}{cccc} \bigcirc & \bigcirc & \bigcirc & \bigcirc \\ \bigcirc & \bigcirc & \bigcirc & \bigcirc \\ \bigcirc & \bigcirc & & \end{array} \right] \div 10 \times 20,000 = \bigcirc$$

$$\text{PLT}(10 \text{ OIF}) \times 20,000 = \text{Estimated PLTs}/\mu\text{L}$$

nRBC WBC Count Correction



$$\text{Corrected WBC} = \frac{(\text{Uncorrected WBC} \times 100)}{(100 + \text{nRBCs})}$$

Worked Example: nRBC Count Correction

Analyzer WBC: $24.8 \times 10^3/\mu\text{L}$ | Differential finds: 18 nRBCs

$$\begin{aligned} & (24.8 \times 100) / (118) \\ & = 2480 / 118 \end{aligned}$$

$21.0 \times 10^3/\mu\text{L}$

Failure to correct = +18% clinical overestimation.

Absolute vs. Relative Leukocyte Concentrations

RELATIVE PERCENTAGE (%)



Percentage of specific leukocyte lineage per 100 counted white blood cells.

ABSOLUTE COUNT (cells/ μ L)



Actual concentration of individual cell lineages in whole blood.

THE ABSOLUTE COUNT FORMULA

$$\text{Total WBC} \times \left(\frac{\text{Relative \%}}{100} \right) = \text{Absolute Cell Count}$$

CLINICAL REALITY CHECK: PERCENTAGES MASK REALITY

DECEPTIVE PRESENTATION (Relative)

70% Lymphs →  **CRITICAL NEUTROPENIA**
in leukopenia

CLINICAL REALITY (Absolute)

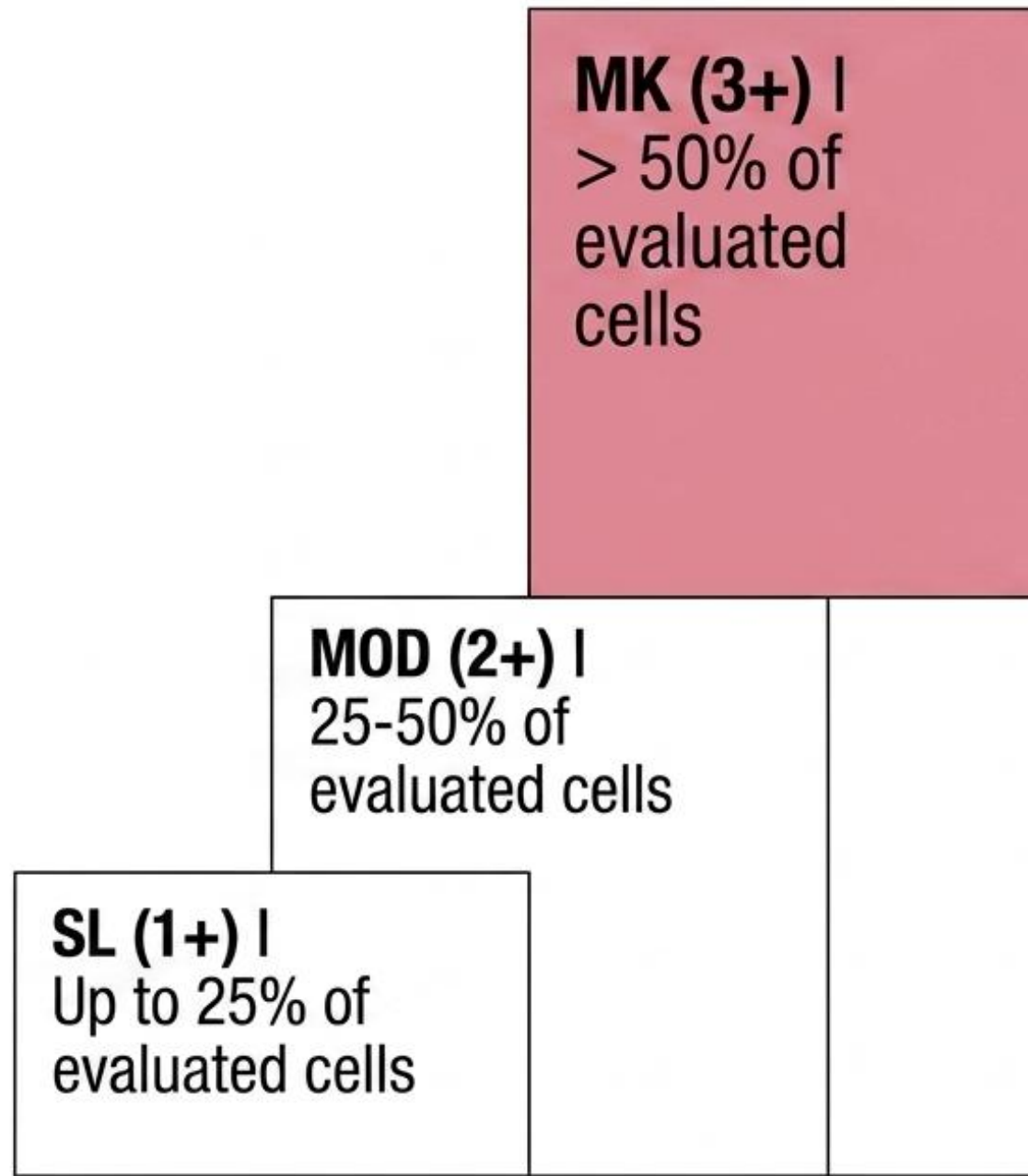
NORMAL ABSOLUTE LYMPHS 

Reference Intervals: Adult Leukocyte Differentials

Lineage	Relative (%)	Absolute ($\times 10^9/L$)
Total WBC	--	4.5 - 11.0
Segs	40 - 70%	1.8 - 7.7
Bands	2 - 8%	0.0 - 0.8
Lymphocytes *	20 - 40%	1.0 - 4.8
Monocytes	2 - 10%	0.2 - 1.0
Eosinophils	1 - 7%	0.05 - 0.5
Basophils	0 - 1%	0.0 - 0.1

* Pediatric Variation (3-12 yrs):
Up to 60% physiologic lymphocyte predominance.

Morphology Grading Tiers



Leukocyte Inclusions

- ☒  Auer Rods
- ☒  Döhle Bodies
- ☒  Toxic Granulation
- ☒  Intracellular Bacteria

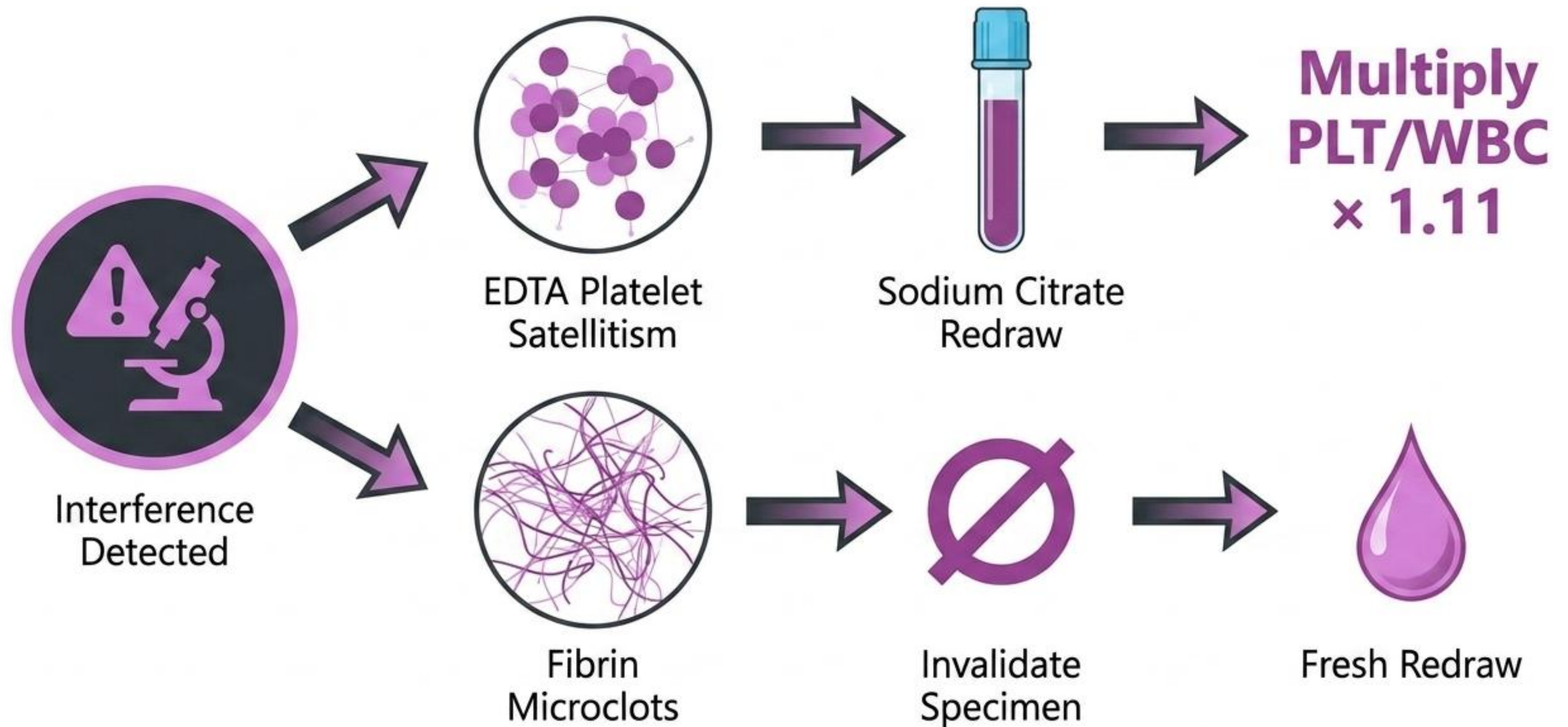
Erythrocyte Inclusions

- ☒  Howell-Jolly Bodies
- ☒  Basophilic Stippling
- ☒  Pappenheimer Bodies
- ☒  Hb C Crystals

Morphology Grading Matrix

	Standard Parameters		
Anisocytosis	SL (1-10%)	MOD (1-20%)	MK (>20%)
Micro/Macrocytosis	SL (1-10%)	MOD (11-20%)	MK (>20%)
Hypochromia	SL (1-10%)	MOD (11-20%)	MK (>20%)
Polychromasia	SL (1-4%)	MOD (5-20%)	MK (>20%)
Unique Poikilocytosis Forms (Schistocytes, Spherocytes)	SL (1-4%)	MOD (5-20%)	MK (>20%)

Pre-Analytical Pitfalls & Technical Interferences



Quality Control & Safety Protocol



CROSS-CHECK CELLULAR ESTIMATES



Verify estimates correlate with analyzer values.



CORRECT FOR nRBC INTERFERENCE

$$\text{nRBC} \geq 5 / 100 \text{ WBC}$$

Use whenever threshold is encountered.



FLAG CRITICAL PATHOLOGIST REVIEW CRITERIA



- ⚠ Blasts
- ⚠ Intracellular Microorganisms
- ⚠ >1% Schistocytes

Immediately escalate smears showing these criteria.



STANDARD UNIVERSAL BIOSAFETY PRECAUTIONS



Use full PPE.



Dispose of all glass slides in rigid sharps biohazard containers.